Staf Bauer

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EDUCATION

PhD, Neuroscience (2022–present)
Erasmus MC, Rotterdam, NL

- Supervisors: Chris de Zeeuw; daily supervision Vincenzo Romano.
- Cerebellar control of locomotion and sensorimotor adaptation; chronic silicon-probe electrophysiology and kinematic tracking in freely moving mice.
- Core strength in quantitative analysis: built a full MATLAB pipeline for spike-kinematics correlation, trajectory clustering, GLM encoding/decoding, and permutation-based statistics.

MSc, Nanobiology (2018–2021)
TU Delft & Erasmus MC, NL

- Thesis (Erasmus MC, lab of Chris de Zeeuw): periodic optogenetic stimulation of the paramedian lobule reveals resonant properties of the cerebellum. Graded 8.5/10; nominated for the Gerrit Jan Mulder Prize.
- Internship at Ciphix (Ciphix Healthcare initiative), graded 9.5/10 (18 ECTS).

BSc, Nanobiology (2013–2018)
TU Delft & Erasmus MC, NL

MBO 2, Security Officer (2016–2017)
IBO / SVPB, Amersfoort, NL

Gymnasium (VWO), NT/NG profile (2007–2013)
Vrijzinnig Christelijk Lyceum, The Hague, NL

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Candidate, Dept. of Neuroscience (2022–present)
Erasmus MC, Rotterdam, NL

- Cerebellar physiology, locomotion, sleep, and adaptive motor control.

RPA Developer / Solution Architect (2018–2022)
Ciphix & Ciphix Healthcare, Rotterdam, NL

- Co-developer of the Ciphix Healthcare initiative (automation of clinical workflows); process analysis, redesign, and automation for KLM, Erasmus MC, VolkerWessels Telecom, Transavia, and Bilfinger.

NOC (Network Operations Centre) (2016–2018)
Workrate, Schiphol-Rijk, NL

- Monitored and processed alarms for EdgeConnex datacenters across Europe (24/7 shift rotation).

PUBLICATIONS

Under revision — Sleep-state dependent cerebellar processing in adult mice
Nature Neuroscience (in revision); preprint: bioRxiv

In preparation — Cerebellar encoding of online gait adjustments
Manuscript in preparation

2026 — The cerebellum engages multiple pre-motor pathways through a divergent-convergent architecture to shape whisker dynamics
Frontiers in Neural Circuits 19 — Co-first author

2025 — Stage-dependent cerebrocerebellar communication during sensorimotor processing
Nature Communications 16, 8812

2024 — Whisker kinematics in the cerebellum
J Physiol 602(1):153–181

2022 — Olivocerebellar control of movement symmetry
Current Biology 32(3):654–670.e4

2022 — Purkinje Cell Activity Resonation Generates Rhythmic Behaviors at 8 Hz
Biomedicines 10(8):1831 — First author; based on MSc thesis

SKILLS

Languages: Dutch (native), English (fluent), Spanish (A2)
Technical: MATLAB, LaTeX, DeepLabCut/Python, MS Office, UiPath (RPA), WordPress
Certificates: Driver's license B (2012); UiPath Developer course (L1–L4)
Interests: Fitness, reading, running, kitesurfing, hockey

LEADERSHIP & EXTRACURRICULAR

Head Coach, Ladies 1 (1e klasse) (2019–2022)
Delftsche Studenten Hockey Club, Delft, NL

- Also: Assistant Coach, Ladies 1 (2018–2019); Head Coach, Ladies 2 (2017–2018).

President (2017–2018)
Delft Student Alpine Club, Delft, NL

- Led a board of 6; organized climbing, skiing, and social activities including tour-skiing events (~150 participants).

Vice President (2015–2016)
Delft Student Hockey Club, Delft, NL

- Responsible for top teams (H1/H2, D1/D2) and coach oversight; secured new club sponsors; co-authored long-term policy for top teams.

Field Hockey — First Team (DSHC) (2013–2018)
Delft, NL

- Team captain (2016–2017); multiple seasons in 1e and 2e klasse.

Inventory Leader — DSC 5-Year Anniversary (2018)
Delft, NL

- Logistics/inventory for a ~3400 visitors/day, 2-week event (3 restaurants, 8 food stands, 14 bars).